AUDITING FACT DELETION IN LANGUAGE MODELS WITH SEPARABLE MEMORY

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ABSTRACT

Language models with separable memory aim to decouple source-specific knowledge from general world knowledge, language understanding and reasoning capabilities. Unlearning then becomes relatively straightforward: simply delete the memory units associated with the targeted knowledge. We show that a naive comparison of model outputs is an unreliable measure of unlearnability, and propose a methodological audit for such systems that measures the effectiveness of memory removal. We apply the audit to Continuous-Query Limited Memory Language Models (Co-LMLM) and Natively Unlearnable Large Language Models (NULLs) on five factual-recall benchmarks. Much of the post-deletion correctness in both systems is also present when separable memory is disabled. This is expected for NULLs by design, but less so for Co-LMLM, which aims to keep facts out of its weights. In NULLs, the backbone recites up to 81.1% of audited facts without any source memory, and removing more sources does not remove more facts but damages up to 13% of nearby ones. For Co-LMLM, up to 21.2% of all facts that pass a naive accuracy test as unlearned are still recited by the backbone. A single entry re-inserted into Co-LMLM's memory restores 37.9–83.0% of facts the model no longer answered after deletion. We propose the audit as a protocol that future work on separable memory can use to support deletion claims.

1 INTRODUCTION

There are two common reasons to delete a fact from a language model. A person may ask for their data to be removed (Fosch-Villaronga et al., 2018), or the training data may contain information that should not have been included. In either case, the operator must both remove the relevant information and provide an auditable basis for the deletion claim (Thudi et al., 2022). This is difficult when knowledge is distributed across shared model parameters. A model may stop producing a fact even though information about it remains internally encoded (Lynch et al., 2024), while removing it from the parameters may require retraining or edits whose broader effects are difficult to control (Bourtoule et al., 2021; Meng et al., 2022; Mitchell et al., 2022a; Cohen et al., 2024). Models with separable memory offer a different approach by placing some knowledge in memory units that can be disabled directly.

Two such systems, Co-LMLM (Feldman et al., 2026) and NULLs (Ghosal et al., 2026), serve as our case studies. Co-LMLM pairs a language-model backbone with an external memory of text entries. NULLs instead associates each training source with a sparse mask over a shared pool of sink neurons, while information common to many sources can remain in the backbone. In both systems, deletion acts on explicit memory units: external entries in Co-LMLM and source-specific memory in NULLs. The central question is whether disabling the units associated with a requested fact is enough to remove that fact from the system.

A standard way to evaluate deletion is to compare the model's output before and after the intervention (Maini et al., 2024; Shi et al., 2025; Li et al., 2024), including in systems with separable memory (Zhao et al., 2026; Feldman et al., 2026; Ghosal et al., 2026). This comparison does not explain why a fact is still recited after deletion. It may be supported by memory units that were not removed, or the backbone may recite it without separable memory at all. The reverse can also happen: the backbone may recite a fact when memory is disabled, while the memory that remains after deletion causes

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the model to produce a different answer. A before-and-after comparison counts this as successful deletion even though the backbone still recites the fact.

We introduce an audit that separates these cases. For each fact, we evaluate the same prompt with the full memory available, with the deletion set disabled while the remaining memory stays active, and with separable memory disabled entirely. Comparing the latter two conditions tells us whether post-deletion recitation depends on memory that survived the deletion or on the backbone, and also reveals cases where surviving memory suppresses an answer the backbone would otherwise produce. We then vary the deletion rule and its breadth to test how reliably different rules identify the memory supporting a fact and what collateral damage broader deletion causes. We apply the same audit to both systems, and add system-specific tests that examine the mechanisms exposed by each architecture.

The audit is intended as a protocol that papers on separable memory can use to support deletion claims, and Section 5 lists what such papers should report.

2 Related Work

Separable memory. Memory-based model editors store each edit outside the pretrained weights, and the memory component holding a given fact is therefore known (Mitchell et al., 2022b; Hartvigsen et al., 2023; Wang et al., 2024). Limited Memory Language Models instead build their memory from a corpus, where the same fact can appear in many entries (Zhao et al., 2026; Feldman et al., 2026). NULLs builds on memorization sinks (Ghosal et al., 2025) and is evaluated on facts taken from the removed source itself (Ghosal et al., 2026), while we start from the fact to be deleted.

When retrieval hurts. Retrieved text can contradict what the model has learned in its weights (Longpre et al., 2021; Xie et al., 2024), and irrelevant text can make a model worse than it would be without retrieval. Mallen et al. (2023) use a fixed retrieval corpus, whereas in our setting, deletion itself can lead the model to retrieve misleading text, which we measure as surviving-memory interference.

Probing and attacks. Linear probes test what can be decoded from a representation (Alain & Bengio, 2017; Hewitt & Liang, 2019), while adversarial passages can hijack retrieval (Zhong et al., 2023) and, in RAG systems, induce attacker-chosen answers (Zou et al., 2025). We use both to test whether a deleted answer remains internally encoded and whether one inserted entry can undo a deletion.

Auditing deletion. Raeesi & Roed (2026) audit deletion by comparing three memory conditions. There, each memory entry is a subject–relation–object triple, and the canonical and alias-equivalent entries for a fact can be enumerated and deleted. We adopt this three-condition design for Co-LMLM and NULLs, whose memory units are free-text entries and training sources rather than triples. We also add surviving-memory interference, deletion rules, and probing and re-insertion tests.

3 Method

3.1 Auditing a deletion

Models with separable memory. We consider language models with *separable memory*, in which part of the model's knowledge resides in components that can be disabled without retraining the rest of the model. Formally, such a system consists of shared parameters θ (the *backbone*) and a memory $\mathcal{M} = \{m_1, \ldots, m_N\}$. We call each $m \in \mathcal{M}$ a *memory unit*. For $\mathcal{M}' \subseteq \mathcal{M}$, we write $p_\theta(\cdot \mid x; \mathcal{M}')$ for the model's distribution over continuations of a prompt x when only the units in $\mathcal{M}'$ are available, and $\hat{y}_\theta(x; \mathcal{M}')$ for the generated continuation.

Facts and recitation. A fact f is given by a prompt x_f and a gold answer a_f. We say that the system *recites* f under memory $\mathcal{M}'$ if the continuation it generates for x_f matches the gold answer:

$$\mathrm{rec}(f; \mathcal{M}') = \mathbf{1}\big[\,\mathrm{match}\big(\hat{y}_\theta(x_f; \mathcal{M}'),\, a_f\big)\big] \in \{0, 1\}, \tag{1}$$

where match is the answer-matching rule given in Section 4.1. We use recitation to assess whether the system possesses the knowledge of a given fact.

Deletion rules and deletion sets. In this audit, a deletion request concerns a fact rather than a particular memory unit. The same fact may be supported by several units, and the backbone may

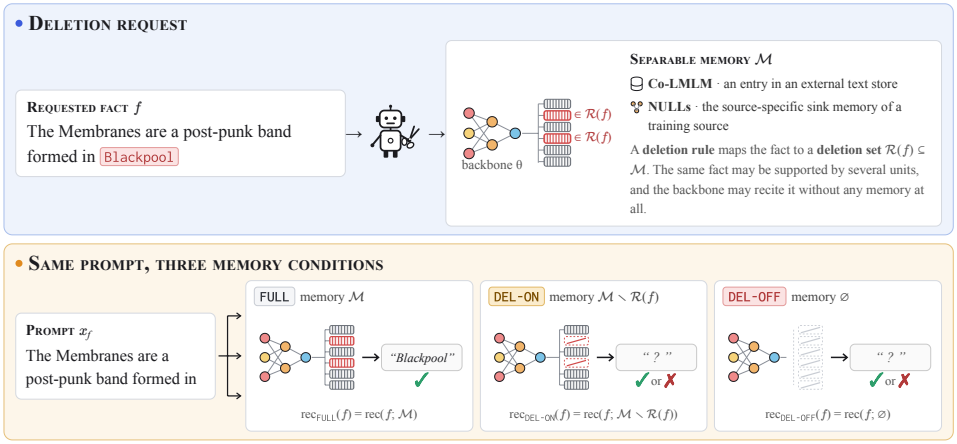


Figure 1: A deletion request and the three memory conditions. *Top:* A deletion rule maps a requested fact f to a deletion set $\mathcal{R}(f) \subseteq \mathcal{M}$ of memory units. *Bottom:* The same prompt x_f is run with the memory unchanged (FULL, memory $\mathcal{M}$), with only the units in $\mathcal{R}(f)$ disabled (DEL-ON, memory $\mathcal{M} \setminus \mathcal{R}(f)$), and with the separable memory disabled entirely (DEL-OFF, memory $\varnothing$).

also be able to answer it without any memory at all. A *deletion rule* $\mathcal{R}$ therefore has to decide which memory units to disable for a requested fact. We write

$$\mathcal{R}(f) \subseteq \mathcal{M}$$

for the resulting set of units, henceforth referred to as the *deletion set*. Different deletion rules can produce different deletion sets for the same fact, so whether a deletion succeeds can depend on both the memory architecture and the rule used to identify what should be removed.

Three memory conditions. To assess the effectiveness of a given deletion set, we evaluate the same prompt under three memory conditions (Figure 1): FULL leaves the memory unchanged; DEL-ON disables the units in $\mathcal{R}(f)$ and leaves the rest of the memory intact; and DEL-OFF disables the separable memory entirely, leaving only the backbone. Formally:

$$\text{rec}_{\text{FULL}}(f) = \text{rec}(f; \mathcal{M}), \qquad \text{memory unchanged}, \tag{2a}$$

$$\text{rec}_{\text{DEL-ON}}(f) = \text{rec}\big(f; \mathcal{M} \setminus \mathcal{R}(f)\big), \qquad \text{deletion set disabled}, \tag{2b}$$

$$\text{rec}_{\text{DEL-OFF}}(f) = \text{rec}(f; \varnothing), \qquad \text{separable memory disabled}. \tag{2c}$$

Only $\text{rec}_{\text{DEL-ON}}$ depends on the deletion rule. $\text{rec}_{\text{DEL-OFF}}$ is a property of the backbone alone. A deletion is only meaningful for a fact that the system recites to begin with, so we audit facts with $\text{rec}_{\text{FULL}}(f) = 1$. We denote the set of audited facts by $\mathcal{F}$.

Separable facts. We say that the system *retains* f if it recites f with its memory intact, $\text{rec}_{\text{FULL}}(f) = 1$. A retained fact is *separable* if the backbone does not recite it:

$$\text{rec}_{\text{FULL}}(f) = 1 \quad \text{and} \quad \text{rec}_{\text{DEL-OFF}}(f) = 0.$$

A retained fact with $\text{rec}_{\text{DEL-OFF}}(f) = 1$ is *non-separable*: the backbone itself recites f, and no memory operation can remove it. Separability is a property of the system $(\theta, \mathcal{M})$ and the prompt x_f, not of the deletion rule.

Audit outcomes. The comparison between DEL-ON and DEL-OFF is the central part of the audit. If the system recites f under both conditions, the fact is not separable. If it recites f under DEL-ON but not under DEL-OFF, the fact is separable, but it is retained through units outside $\mathcal{R}(f)$, which means the deletion set is incomplete. The reverse case is also important. If the system recites f under DEL-OFF but not under DEL-ON, the backbone can recite the fact, and the surviving memory $\mathcal{M} \setminus \mathcal{R}(f)$ suppresses an answer the backbone would otherwise produce.

We summarize these outcomes with four per-fact quantities:

$$\mathrm{PDR}(f) = \mathrm{rec}_{\text{DEL-ON}}(f), \qquad \text{post-deletion recitation,} \tag{3a}$$
$$\mathrm{BBR}(f) = \mathrm{rec}_{\text{DEL-OFF}}(f), \qquad \text{backbone recitation,} \tag{3b}$$
$$\mathrm{SMS}(f) = \mathrm{rec}_{\text{DEL-ON}}(f)\,\big(1 - \mathrm{rec}_{\text{DEL-OFF}}(f)\big), \qquad \text{surviving-memory support,} \tag{3c}$$
$$\mathrm{SMI}(f) = \big(1 - \mathrm{rec}_{\text{DEL-ON}}(f)\big)\,\mathrm{rec}_{\text{DEL-OFF}}(f), \qquad \text{surviving-memory interference.} \tag{3d}$$

Applied to a set of facts, each quantity denotes its mean over that set, for example $\mathrm{PDR}(\mathcal{F}) = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}}\mathrm{PDR}(f)$. We drop the argument when the set is the audited cohort $\mathcal{F}$, so that PDR, BBR, SMS, and SMI without an argument denote rates.

Post-deletion recitation (PDR) is what a standard before-and-after evaluation observes. The other quantities explain why that outcome occurred. Surviving-memory support (SMS) identifies facts that are recited only because of memory the deletion left behind, while surviving-memory interference (SMI) identifies facts that appear deleted only because the remaining memory causes an error. A before-and-after test counts every fact with $\mathrm{SMI}(f) = 1$ as successfully deleted, even though the backbone still recites it. To make this failure directly interpretable, we also report the *spurious-deletion rate*:

$$\Phi = \frac{\mathrm{SMI}}{1 - \mathrm{PDR}}. \tag{4}$$

Since every audited fact is recited under FULL, $1-\mathrm{PDR}$ is the fraction of facts that a before-and-after test counts as deleted, and Φ is the fraction of these that the backbone still recites.

Varying deletion breadth. Many deletion rules possess a breadth parameter b that controls how many units are disabled by them. Disabling more units might make the target harder to recite, but it might also affect other facts, so we measure both effects. Let $\mathcal{N}(f)$ be a set of nearby control facts, with $f \notin \mathcal{N}(f)$. Let $\mathcal{R}_b(f)$ be the deletion set that the rule produces for f at breadth b. Each control fact is evaluated under the deletion constructed for the target f, that is, with memory $\mathcal{M} \setminus \mathcal{R}_b(f)$:

$$E(f,b) = 1 - \mathrm{rec}\big(f; \mathcal{M} \setminus \mathcal{R}_b(f)\big), \qquad \text{target removal,} \tag{5a}$$
$$X(f,b) = \frac{\sum_{f'\in\mathcal{N}(f)} \mathrm{rec}_{\text{FULL}}(f')\,\big(1 - \mathrm{rec}(f'; \mathcal{M} \setminus \mathcal{R}_b(f))\big)}{\sum_{f'\in\mathcal{N}(f)} \mathrm{rec}_{\text{FULL}}(f')}, \qquad \text{collateral damage.} \tag{5b}$$

Here, $E(f,b) = 1$ when the target is no longer recited, and the collateral $X(f,b)$ is the fraction of control facts that were recited before the deletion and are no longer recited afterward.

3.2 Applying the audit to Co-LMLM and NULLs

We apply the same audit to two systems with different forms of separable memory: Co-LMLM (Feldman et al., 2026) and NULLs (Ghosal et al., 2026). The systems differ in what a memory unit is, in how the model reads from an enabled memory $\mathcal{M}'$, and in the rules used to construct $\mathcal{R}(f)$.

Co-LMLM. In Co-LMLM, the backbone is a language model with parameters θ, and the separable memory is an external key–value store whose entries are the memory units: $\mathcal{M} = \{(k_i, v_i)\}_{i=1}^{N}$. Each key k_i is an L_2-normalized hidden state emitted at a `<FACT>` token, and each value v_i is the associated text span. When the model emits `<FACT>`, its L_2-normalized hidden state q queries the enabled memory $\mathcal{M}'$ by cosine similarity,

$$(k^\star, v^\star) = \arg\max_{(k,v)\in\mathcal{M}'} \langle q, k\rangle.$$

If $\langle q, k^\star\rangle > \tau$ for the retrieval threshold τ, the text $v^\star$ is inserted into the generation context; otherwise, and always when $\mathcal{M}' = \varnothing$, generation continues without retrieval.

The three conditions of Equation 2 amount to retrieving from the full store (FULL), filtering the entries in $\mathcal{R}(f)$ while retrieving from the rest (DEL-ON), and disabling retrieval (DEL-OFF). The difficulty is deciding which entries belong in $\mathcal{R}(f)$. Let $\hat{q}_f$ be the query formed for x_f under FULL, and let $m_f^\star \in \mathcal{M}$ be the entry it retrieves. We introduce several rules. Every rule disables $m_f^\star$, which serves as an oracle seed, together with a set of related entries: the *geometric* rule $\mathcal{R}_\rho^{\text{geo}}$ adds entries whose keys have cosine similarity at least ρ with $\hat{q}_f$; the *value* rule $\mathcal{R}^{\text{val}}$ adds entries whose text

mentions a_f; the *provenance* rule $\mathcal{R}^{\text{prov}}$ adds entries from the same source document as $m_f^\star$; and the <FACT-q> rule $\mathcal{R}^{\texttt{FACT-q}}$ generates alternative questions about the same fact and adds the entries retrieved by those questions. Appendix D.1 gives the formal definitions.

Unless stated otherwise, the standard Co-LMLM deletion is the union of the geometric and value rules, $\mathcal{R}^{\text{geo}}_{0.85} \cup \mathcal{R}^{\text{val}}$. We choose $\rho = 0.85$ as a middle ground within the range we later sweep.

NULLs. NULLs stores its separable memory as trainable parameters. Each training source s is assigned a sparse mask μ_s over a shared pool of sink neurons. Activating μ_s determines which of these neurons participate in the forward pass, while knowledge shared across many sources can remain in the backbone. The memory units are the sources, $\mathcal{M} = \mathcal{S}$, and disabling a source removes its mask from those the model can use.

A deletion request names a fact, not a source. We therefore associate each fact with the source whose Wikipedia article title matches the subject of the fact. When matching, we ignore underscores versus spaces, extra whitespace, and the case of the first character. We only audit facts that have a matching title. FULL uses the mask of this source. NULLs activates one source mask per forward pass (Ghosal et al., 2026). Under DEL-ON, we activate the mask of the source whose title embedding is closest to that of the deleted source. Under DEL-OFF, all sink neurons are disabled, leaving only the shared backbone. Since NULLs is never trained with all sinks disabled, Appendix C also tests a DEL-OFF that instead activates the mask of an unrelated source.

The standard NULLs deletion rule is the provenance rule $\mathcal{R}^{\text{prov}}$, which deletes only the source associated with the fact. We also test broader source-level rules: the geometric rule $\mathcal{R}^{\text{geo}}_k$ adds the k nearest source articles under an external text encoder, the value rule $\mathcal{R}^{\text{val}}$ adds nearby sources whose article text mentions the answer, and their combination $\mathcal{R}^{\text{geo}}_k \cap \mathcal{R}^{\text{val}}$ adds sources that satisfy both criteria. Larger k removes more sources and thus broadens deletion. Nearby source representations also define the control facts used to measure collateral damage. The formal definitions are in Appendix D.2.

We only audit facts where the memory to be deleted supports the fact, since deleting memory that does not support it cannot test whether deletion works. For Co-LMLM, the entry retrieved under FULL must mention the gold answer a_f. For NULLs, the mask of the associated source must make a_f more likely than the mask of an unrelated source. This may exclude facts the backbone has fully memorized, which would make deletion look more effective than it is. Our estimates of backbone recitation for NULLs are therefore conservative.

Co-LMLM and NULLs differ in model size, training data, objective, and memory implementation. We therefore treat them as two applications of the same audit rather than as a controlled comparison between architectures.

System-specific tests. Some questions require interfaces that only one of the systems exposes. For Co-LMLM, we use the contents of the retrieval store to ask whether answers that occur more densely near a query are also more likely to be recited by the backbone. We use a linear probe to ask whether answer information can be decoded from the model's query representation even when the backbone does not recite that answer under DEL-OFF. Finally, we test whether placing a single answer-bearing entry just outside a geometric deletion boundary can reverse a deletion. Appendix E gives the full definitions. For NULLs, we ask more directly how much the source-specific sink contributes to the fact. We compare the answer likelihood under the associated source mask and an unrelated source mask, measure how often turning the associated source on or off changes the generated answer, and replace the next-closest surviving source with an unrelated source to test whether post-deletion behavior depends on which source remains active. Appendix F gives the full definitions.

4 Results and Discussion

4.1 Experimental setup

Datasets. We evaluate on five factual-recall datasets: T-REx (Elsahar et al., 2018), PopQA (Mallen et al., 2023), Google-RE (Petroni et al., 2019), CounterFact (Meng et al., 2022), and ZsRE (Levy et al., 2017). Each example specifies a fact, its gold answer, and a set of accepted aliases. T-REx,

Table 1: Rates of Equation 3 and 4 under each system's standard deletion rule, in percent of the audited cohort $\mathcal{F}$, $n = |\mathcal{F}|$; ↓ marks rates where lower means more complete deletion. Reference models have no separable memory; their rows give accuracy on the same facts.

	T-REx	Google-RE	CounterFact
Co-LMLM			
n	7,407	608	3,479
Post-deletion recitation PDR ↓	8.3	0.3	5.9
Backbone recitation BBR ↓	25.6	0.2	24.2
Surviving-memory support SMS ↓	1.5	0.2	1.6
Surviving-memory interference SMI ↓	18.8	0.0	19.9
Spurious-deletion rate Φ ↓	20.5	0.0	21.2
SmolLM2-360M	41.3	5.8	43.6
Std-LM-360M	46.8	3.0	46.1
NULLs			
n	2,669	35	791
Post-deletion recitation PDR ↓	80.7	54.3	67.1
Backbone recitation BBR ↓	81.1	62.9	61.6
Surviving-memory support SMS ↓	8.1	8.6	17.3
Surviving-memory interference SMI ↓	8.4	17.1	11.8
Spurious-deletion rate Φ ↓	43.7	37.5	35.8
SmolLM2-1.7B	82.2	40.0	64.7

Google-RE, and CounterFact are evaluated with cloze-style prompts, while PopQA and ZsRE use short-answer questions. Appendix B describes how we construct each prompt set and gives examples.

Answer matching. For generated outputs, we count an answer as correct when the model produces the gold answer or one of its aliases as a complete phrase after casefolding and normalizing punctuation and whitespace. For Co-LMLM memory entries, we use a slightly more permissive rule because an entry may contain a shorter surface form of the same answer. We define an entry as answer-relevant if it contains the gold answer a_f or an accepted alias, or if the entry itself is a sufficiently long phrase within an alias, such as "Oxford" in "University of Oxford". This rule is used in the Co-LMLM pre-deletion check (Section 3.2) and the value rule (Appendix D.1).

Selecting facts for evaluation. The deletion audit is meaningful only when the separable memory being tested is relevant to the fact before deletion. We therefore apply a system-specific pre-deletion check. The audited cohort $\mathcal{F}$ consists of the facts that pass the pre-deletion check (Section 3.2) and are recited under FULL. We report FULL accuracy over all facts that pass the check. Because the two systems use different checks, each has its own evaluation set.

Reference models and decoding. To put backbone recitation in context, we also evaluate conventional language models without separable memory. For Co-LMLM, we use its matched 360M standard-LM baseline and SmolLM2-360M. For NULLs, we use SmolLM2-1.7B. These models do not undergo deletion; their accuracy simply shows how often the same facts are answered without a separable memory mechanism. All models use greedy decoding with at most 12 new tokens. Appendix H gives the checkpoints and remaining evaluation settings.

4.2 What supports an answer after deletion?

Our first experiment asks why a fact is still recited after deletion. For each system, we apply its standard deletion rule and compare DEL-ON with DEL-OFF. If a fact is recited only under DEL-ON, it depends on memory that survived (SMS). If it is also recited under DEL-OFF, the backbone recites it without that memory. Table 1 reports these outcomes.

On T-REx, Co-LMLM recites 8.3% of facts under DEL-ON (PDR), but only 1.5% are recited because of surviving memory (SMS). For NULLs, the corresponding rates are 80.7% and 8.1%. The same pattern appears on CounterFact. Therefore, most of the facts recited after deletion are

answerable by the backbone alone. Deleting more memory cannot remove them; whether they can be deleted depends on how well training keeps them out of the backbone.

Deletion can also produce the opposite effect. Some facts are no longer recited under DEL-ON even though the backbone recites them under DEL-OFF. This surviving-memory interference (SMI) affects 18.8% of Co-LMLM facts on T-REx and 19.9% on CounterFact. NULLs also shows interference, at 8.4% on T-REx and 11.8% on CounterFact. The later system-specific analyses of NULLs suggest that many of these cases involve facts supported mainly by the shared backbone, while changing the active source mask can still perturb the generated answer. In Co-LMLM, interference can instead arise when the remaining external memory retrieves text that overrides an answer the backbone recites without retrieval.

For Co-LMLM, the spurious-deletion rate Φ is 20.5% on T-REx and 21.2% on CounterFact. For NULLs, Φ is 43.7% on T-REx and 35.8% on CounterFact. These rates show how often a before-and-after comparison can give a misleading picture of what deletion changed. For Co-LMLM, the backbone still recites the fact, but retrieval after deletion hides it. For NULLs, this mostly reflects deletion changing recitation about equally often in both directions ($\mathrm{SMS} \approx \mathrm{SMI}$), not facts hidden by the remaining memory (see Appendix F).

The backbone recitation BBR is below the accuracy of both 360M reference models on every prompt set for Co-LMLM. Externalizing factual text therefore reduces how often the backbone recites these facts, but does not eliminate it.

Google-RE is an exception for Co-LMLM, where post-deletion and backbone recitation are both near zero. The reference models, however, also answer only 3.0–5.8% of these facts, which suggests that the result reflects the difficulty of the prompt set more than the deletion. For NULLs, the Google-RE cohort ($n = 35$) is too small to draw conclusions.

For space, Appendix G reports PopQA and ZsRE. In Co-LMLM, surviving-memory support stays below 1% on both. PopQA resembles T-REx and CounterFact, with interference far exceeding support, while ZsRE resembles Google-RE, with all rates near zero. In NULLs, post-deletion and backbone recitation remain close, though both cohorts are small ($n = 22$ and 79).

4.3 Does the deletion rule matter?

The previous experiment uses one standard deletion rule for each system. We next ask whether the outcome changes when the same fact is mapped to different deletion sets. For Co-LMLM, we compare the oracle seed, geometric, provenance, <FACT-q>, and value rules described in Section 3.2, together with their unions. For NULLs, we compare the provenance, geometric, and value rules, and the intersection of the latter two. Table 12 reports T-REx, Google-RE, and CounterFact.

For Co-LMLM, the value rule is much more effective at removing supporting memory because it explicitly uses the answer text when constructing the deletion set. The geometric, provenance, and <FACT-q> rules leave surviving-memory support (SMS) close to that of removing only the retrieved entry (30.0–36.3% on T-REx, 41.7–46.1% on CounterFact). However, every rule that includes the value rule reduces SMS to below 2%. The entries that support the answer therefore lie in other source documents and below the geometric threshold. In short, explicitly using the answer text when constructing the deletion set identifies supporting entries that the other rules miss.

This reduction comes at a cost. Surviving-memory interference rises from 6.7 to 18.8% on T-REx and from 5.7 to 19.9% on CounterFact. A more complete deletion therefore makes more facts appear deleted, even though the backbone still recites them.

On Google-RE, the value rule likewise reduces surviving-memory support (SMS) from 14.1 to 0.2%, but without raising interference. The backbone recites almost none of these facts, meaning there is almost nothing for the remaining memory to hide.

For NULLs, the choice of rule barely changes surviving-memory support (SMS). This is consistent with Appendix F, where SMS is unchanged when the next-closest source is replaced by an unrelated one: support after deletion is not tied to any particular source, leaving broader rules little to remove. The value rule instead roughly halves surviving-memory interference (SMI) on T-REx and CounterFact. The Google-RE cohort ($n = 35$) is too small to draw conclusions.

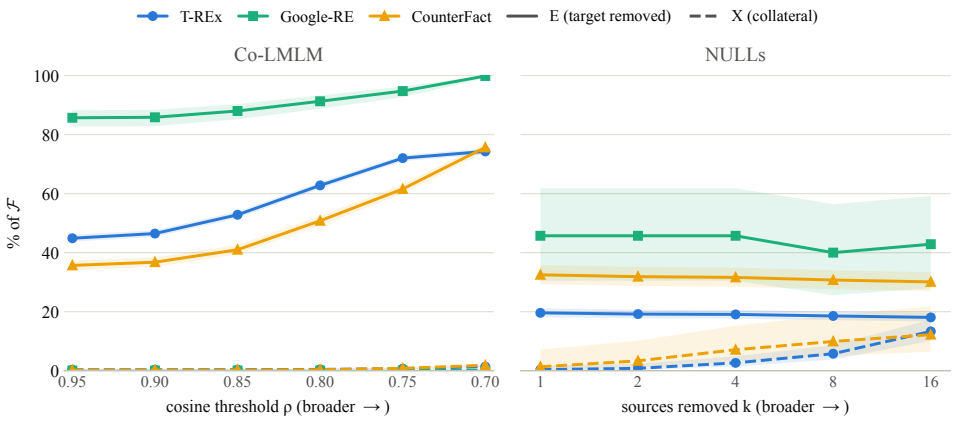


Figure 2: Effect of increasing deletion breadth. E is the fraction of target facts no longer recited after deletion, and X is the fraction of nearby facts recited under FULL that are no longer recited after applying the target's deletion. Both are in percent, averaged over the targets in the audited cohort $\mathcal{F}$. Intervals are Wilson 95% intervals (16). PopQA and ZsRE are reported in Figure 7. For NULLs, Google-RE had too few control facts for X to be plotted.

For space, Appendix G reports the remaining datasets. On PopQA and ZsRE, only rules that include the value rule bring surviving-memory support in Co-LMLM to about 1%, at the cost of more interference on PopQA. In NULLs, the value rule lowers surviving-memory support, but the cohorts are too small to draw conclusions.

4.4 What happens when deletion is made broader?

We now hold the geometric rule ($\mathcal{R}^{\text{geo}}$) fixed and vary its breadth. This asks whether disabling more memory units makes the target fact less likely to be recited, and whether doing so also affects nearby facts. For Co-LMLM, lowering the cosine threshold ρ broadens geometric deletion. For NULLs, increasing k adds more neighboring sources to the deletion set. Since ρ and k control different operations, we examine the trend within each system rather than compare their values directly.

Although ρ and k are not comparable, the two systems respond to breadth in opposite ways. In Co-LMLM, broader deletion removes more targets at little cost to nearby facts, whereas in NULLs it costs nearby facts without removing more targets. This reflects where the audited facts mainly reside, in separable memory for Co-LMLM and in the shared backbone for NULLs.

4.5 Robustness checks

Surviving-memory support and interference both depend on how DEL-OFF is implemented, because DEL-OFF determines whether the backbone recites the fact. We therefore test a second implementation of DEL-OFF for each system. For NULLs, we also vary the pre-deletion source check used to select facts for evaluation. Appendix C reports the full results.

For Co-LMLM, the two implementations of DEL-OFF give nearly the same result. Backbone recitation differs by at most 1.0 percentage points across prompt sets, and surviving-memory interference differs by at most 0.3 points. The interference we observe therefore does not depend on the particular way retrieval is disabled.

NULLs is more sensitive to the choice of DEL-OFF, especially on the smallest evaluation sets, so we treat its exact rates of surviving-memory support and interference more cautiously. We also tighten the pre-deletion verification requirement so that the associated source must support the gold answer by a larger margin. This removes many examples, but backbone recitation remains 69.7% on T-REx and 44.3% on CounterFact at the strictest tested margin. Even among facts most clearly supported by their associated source, the backbone recites many of them without source-specific sink memory.

4.6 SYSTEM-SPECIFIC ANALYSES

The experiments above use the same basic audit for both systems. We finish with additional tests that use interfaces specific to each architecture.

We first ask whether facts whose answers occur more densely near the query in the external store are also more likely to be recited by the backbone. Answer density is positively associated with backbone recitation, but less strongly for Co-LMLM than for the closed-book references. We next ask whether answer information can be decoded from Co-LMLM's query representation even when the backbone does not recite the answer under DEL-OFF. A linear probe exceeds the strongest control by 5.6 points on T-REx, 10.0 on CounterFact, and 4.4 on Google-RE; the evidence is weak on PopQA and absent on ZsRE. Finally, we test whether one surviving memory entry can reverse a deletion. A single verbatim answer-bearing entry placed just outside the geometric deletion boundary restores recitation for 37.9–83.0% of facts that were not recited after geometric deletion. Appendix E gives the full analyses. For NULLs, we ask how much the source-specific sink associated with a fact contributes to the model's behavior. On four of the five prompt sets, the associated source mask makes the gold answer more likely than an unrelated source mask for only 50–52% of facts, and switching the associated source mask on or off changes fewer than 5% of generated answers in each direction. We then replace the next-closest source used after deletion with an unrelated source. The resulting rates of surviving-memory support and interference change little on the larger evaluation sets. Together, these tests suggest that the evaluated facts are supported mainly by the shared backbone rather than by one particular source-specific sink, which is consistent with the intended design of NULLs (Ghosal et al., 2026). Appendix F gives the full analyses.

5 CONCLUSION

We introduced an audit that evaluates each fact with memory intact, with the deletion set disabled, and with separable memory disabled, and applied it to Co-LMLM and NULLs across five factual-recall benchmarks. In Co-LMLM, deletion appears more successful than it is. Up to 21.2% of facts that a before-and-after comparison counts as deleted are still recited by the backbone, hidden only by the remaining memory, and a single similar entry added outside the deletion set restores up to 83.0% of deleted facts. In NULLs, the shared backbone recites up to 81.1% of audited facts without any source memory, consistent with its design for knowledge shared across sources. For such facts, deleting more sources does not remove more targets but damages up to 13% of nearby facts.

Our findings suggest what papers on separable memory should report to support a deletion claim. First, they should define how a requested fact is mapped to memory units, since the choice of rule largely determines how much supporting memory survives. Second, they should evaluate each fact under all three memory conditions and report the spurious-deletion rate Φ, since a before-and-after comparison alone counts facts as deleted that the backbone still recites. Third, they should vary deletion breadth, since broader deletion can remove more targets or only harm nearby facts, depending on where facts reside. Finally, they should test how DEL-OFF is implemented, and, for external memory, whether a single added entry can undo a deletion.

The results also point to one possible change for each system. Co-LMLM inserts the nearest entry above its retrieval threshold without checking whether it concerns the queried fact. This may be part of what allows the remaining memory to hide answers after deletion, and a single inserted entry to disrupt it. A learned check on retrieved entries could make what the backbone still recites visible rather than hidden. In NULLs, facts shared across articles appear to be held by the backbone because no single source sink holds them. Assigning each document an entity- or cluster-level sink alongside its article sink might give such facts a unit that source deletion can remove.

Limitations. Recitation is measured with greedy decoding and answer matching, and backbone recitation is therefore a lower bound on what the backbone retains, as the representation probe indicates. We audit one model size per system and a fixed set of deletion rules, and other rules may remove supporting memory with less interference. For NULLs, we use standard factual-recall benchmarks rather than facts drawn from its training articles, which yields small audited cohorts on several of the benchmarks. A matched baseline for each system would allow comparing how much Co-LMLM and NULLs each reduce backbone recitation relative to a model without separable memory.

AI USE STATEMENT

In this work, we used generative AI tools to debug code and to improve the clarity of the writing. We have reviewed all AI-assisted work and take responsibility for the final content of this work.

ETHICS STATEMENT

This work audits deletion in publicly released models, memory stores, and prompt sets, and involves no human subjects. The audited data contain facts about real people, but we construct no new records about individuals, and our released outputs contain only benchmark prompts, model outputs, and memory entries already present in the public store.

The adversarial re-insertion test and the representation probe are dual-use. Both require privileged access, write access to the memory store or read access to internal representations, and both target the operator's deletion procedure rather than any individual.

Our results concern whether deletion removes a fact from model behavior and representations. They do not establish whether any system meets a legal standard for erasure.

REPRODUCIBILITY STATEMENT

All audited models, memory stores, and prompt sets are publicly released. Appendix H lists the exact checkpoints, settings, and hyperparameters for every run. Appendix B describes how the prompt sets and audited cohorts are built. Appendices D, E, and F give the formal definitions of the deletion rules and all additional analyses. Appendix I describes the confidence intervals and statistical tests. Decoding is greedy with a fixed global seed. Appendix A reports the compute required.

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A Compute and Reproducibility

Jobs. The full audit consists of 185 jobs: 95 for Co-LMLM, 75 for NULLs, and 5 for each of the three reference models. The jobs are created and distributed by a scheduler in our code.

GPU hours. Co-LMLM took 110 GPU hours, and its two 360M reference models took 19 and 18 GPU hours. The NULLs audit took 127 GPU hours and the SmolLM2-1.7B reference took 22 GPU hours. In total, the audit took 296 GPU hours.

Models and data. We use the released Co-LMLM memory store built from FineWeb-Edu and Wikipedia (Feldman et al., 2026), indexed with FAISS (Johnson et al., 2021). It takes 1.07 TB of disk space. For NULLs, we use the released Wikipedia checkpoint, which takes 12.08 GB, and its released mapping from 6,407,814 article titles to source identifiers, which takes 179.3 MB.

B Dataset and Cohort Construction

We build all five prompt sets with released preparation scripts. Each fact has one prompt, a canonical gold answer, and a set of aliases.

T-REx. We use cloze prompts from the LAMA release (Petroni et al., 2019). Each prompt is the shortest prefix of the source sentence that contains the subject and is at least 30 characters long. The gold answer is the answer as it appears in that sentence, and the object label from the knowledge base is an alias.

PopQA. We use questions from the test split, restricted to subjects with a Wikipedia popularity of at least 10^4. The first listed answer is gold, and the rest are aliases.

Google-RE. We use the born-in and died-in relations from LAMA as template cloze prompts, excluding birth dates and keeping at most 2,000 facts per relation.

CounterFact. We fill the rewrite prompt with the subject and score it against the true answer, not the counterfactual one.

ZsRE. We use the evaluation release from MEND (Mitchell et al., 2022a). The first listed answer is gold, and the rest are aliases.

Table 2: Three example prompts per set, shown verbatim. For PopQA and ZsRE, the suffix "The answer is" is omitted. The gold column shows the canonical answer.

Dataset	Prompt	Gold
T-REx	The Membranes are a post-punk band formed in	Blackpool
	Agaléga (French: îles Agaléga) are two Outer islands of	Mauritius
	Palacio Municipal de Deportes de Granada is an arena in	Granada
PopQA	What genre is Sonic Youth?	alternative rock
	Who is the mother of Alexander the Great?	Olympias
	What is the capital of Trinidad and Tobago?	Port of Spain
Google-RE	Jacob Henry Studer was born in	Columbus
	Georgina Willis was born in	Australia
	Stanley Corrsin was born in	Philadelphia
CounterFact	Autonomous University of Madrid, which is located in	Spain
	The mother tongue of Thomas Joannes Stieltjes is	Dutch
	Porsche 911, created by	Porsche
ZsRE	Who was the designer of Lahti Town Hall?	Eliel Saarinen
	What architect designed Toodyay Fire Station?	Ken Duncan
	What is the name of Nebaioth father?	Ishmael

C Robustness Checks

C.1 Alternative DEL-OFF implementation

Surviving-memory support SMS and surviving-memory interference SMI are read off a comparison with DEL-OFF, and therefore both depend on how memory is disabled. Table 3 repeats the standard deletion rule under each system's second implementation of DEL-OFF, with the cohort and the deletion rule unchanged. This tells us whether the decomposition describes the system or the particular way we disabled its memory.

For Co-LMLM, the standard implementation, NULL-RETRIEVAL, lets the model emit <FACT> but filters every retrieval candidate, so generation continues without inserted text. The alternative, FORBID-TOKEN, prevents <FACT> from being emitted at all, so the model never attempts retrieval. For NULLs, the standard implementation, SINKS-ZERO, deactivates all sink neurons, while the alternative, PLACEBO-SINK, instead activates the mask of an unrelated source.

Table 3: Outcomes under two implementations of DEL-OFF, in percent of the audited cohort $\mathcal{F}$, with Wilson 95% intervals (16). In Co-LMLM, NULL-RETRIEVAL filters every candidate and FORBID-TOKEN prevents <FACT> emission entirely. In NULLs, SINKS-ZERO deactivates the source-specific sink neurons and PLACEBO-SINK activates the mask of an unrelated source in their place. Each implementation label refers to the rate in the row above it and the interval in its own row.

	DEL-OFF	T-REx	Google-RE	CounterFact	PopQA	ZsRE
Co-LMLM						
n		7,407	608	3,479	761	2,260
BBR	NULL-RETRIEVAL	25.6 [24.6, 26.6]	0.2 [0.0, 0.9]	24.2 [22.8, 25.7]	8.4 [6.6, 10.6]	0.9 [0.6, 1.4]
	FORBID-TOKEN	24.6 [23.7, 25.6]	0.3 [0.1, 1.2]	23.6 [22.2, 25.0]	8.7 [6.9, 10.9]	0.9 [0.6, 1.4]
SMS	NULL-RETRIEVAL	1.5 [1.2, 1.8]	0.2 [0.0, 0.9]	1.6 [1.2, 2.1]	0.9 [0.4, 1.9]	0.7 [0.4, 1.1]
	FORBID-TOKEN	2.5 [2.2, 2.9]	0.2 [0.0, 0.9]	2.2 [1.8, 2.8]	0.9 [0.4, 1.9]	0.7 [0.4, 1.1]
SMI	NULL-RETRIEVAL	18.8 [17.9, 19.7]	0.0 [0.0, 0.6]	19.9 [18.7, 21.3]	8.1 [6.4, 10.3]	0.7 [0.4, 1.1]
	FORBID-TOKEN	18.8 [18.0, 19.8]	0.2 [0.0, 0.9]	19.9 [18.7, 21.3]	8.4 [6.6, 10.6]	0.8 [0.5, 1.2]
NULLs						
n		2,669	35	791	22	79
BBR	SINKS-ZERO	81.1 [79.5, 82.5]	62.9 [46.3, 76.8]	61.6 [58.1, 64.9]	72.7 [51.8, 86.8]	38.0 [28.1, 49.0]
	PLACEBO-SINK	79.8 [78.3, 81.3]	57.1 [40.9, 72.0]	67.6 [64.3, 70.8]	36.4 [19.7, 57.0]	38.0 [28.1, 49.0]
SMS	SINKS-ZERO	8.1 [7.1, 9.1]	8.6 [3.0, 22.4]	17.3 [14.8, 20.1]	13.6 [4.7, 33.3]	22.8 [14.9, 33.2]
	PLACEBO-SINK	8.3 [7.3, 9.4]	17.1 [8.1, 32.7]	13.8 [11.6, 16.4]	40.9 [23.3, 61.3]	19.0 [11.9, 29.0]
SMI	SINKS-ZERO	8.4 [7.4, 9.5]	17.1 [8.1, 32.7]	11.8 [9.7, 14.2]	13.6 [4.7, 33.3]	20.3 [12.9, 30.4]
	PLACEBO-SINK	7.5 [6.5, 8.5]	20.0 [10.0, 35.9]	14.3 [12.0, 16.9]	4.5 [0.8, 21.8]	16.5 [9.9, 26.1]

For Co-LMLM, the choice of implementation barely moves the decomposition. The two estimates of backbone recitation BBR differ by at most 1.0 points, SMI by at most 0.3, and SMS stays small under both at no more than 2.5%. We read this as evidence that surviving-memory interference SMI comes from what Co-LMLM retrieves after deletion, since it is almost identical under both implementations, while what movement there is sits mostly in BBR and SMS.

The picture is different for NULLs. The largest cohort, T-REx, remains stable, but BBR moves by 6.1 points on CounterFact. Additionally, the three small cohorts move much further. For instance,

on PopQA ($n = 22$), SMS rises from 13.6 to 40.9% while SMI falls from 13.6 to 4.5%. Since the largest movements fall on the smallest cohorts, we cannot separate genuine sensitivity to the implementation from sampling noise. Therefore, we read the NULLs values of SMS and SMI as indicative rather than exact.

C.2 Stricter source verification

Before auditing a fact in NULLs, we check that its associated source actually supports it. The associated-source mask, the sink mask assigned to that source article, must make the answer more likely than a placebo mask, the mask of a source deterministically chosen to be unrelated to it. We accept any positive difference, but that difference is usually small, with a median of 0.010 nats on T-REx (see Table 7). The cohort therefore contains many facts that their own source barely supports. Table 4 raises the required difference δ, keeping only facts whose source clearly supports them, and recomputes the outcomes on what remains. The question is whether our results hold on the facts where deleting the source should matter most.

Table 4: NULLs outcomes under the standard deletion rule as the verification-gate margin δ is tightened, in percent of the audited cohort $\mathcal{F}$, with Wilson 95% intervals. "Passing" is the number of facts that pass the gate and $n = |\mathcal{F}|$ the subset of those recited under FULL. The row marked ‡ is the threshold used for the results reported elsewhere in the paper.

δ	Passing	n	PDR	BBR	SMS	SMI
T-REx						
0.00‡	10,711	2,669	80.7 [79.2, 82.2]	81.1 [79.5, 82.5]	8.1 [7.1, 9.1]	8.4 [7.4, 9.5]
0.25	6,468	1,329	72.2 [69.8, 74.6]	74.3 [71.9, 76.6]	9.0 [7.6, 10.7]	11.1 [9.6, 12.9]
0.50	3,749	672	66.7 [63.0, 70.1]	71.3 [67.7, 74.6]	8.5 [6.6, 10.8]	13.1 [10.8, 15.9]
1.00	1,309	221	61.5 [55.0, 67.7]	69.7 [63.3, 75.4]	8.6 [5.6, 13.0]	16.7 [12.4, 22.2]
CounterFact						
0.00‡	8,915	791	67.1 [63.8, 70.3]	61.6 [58.1, 64.9]	17.3 [14.8, 20.1]	11.8 [9.7, 14.2]
0.25	6,222	516	62.4 [58.1, 66.5]	57.0 [52.7, 61.2]	18.8 [15.7, 22.4]	13.4 [10.7, 16.6]
0.50	4,011	326	56.7 [51.3, 62.0]	52.1 [46.7, 57.5]	18.7 [14.9, 23.3]	14.1 [10.7, 18.3]
1.00	1,561	115	47.0 [38.1, 56.0]	44.3 [35.6, 53.5]	21.7 [15.2, 30.1]	19.1 [13.0, 27.3]

At $\delta = 1.00$, 1,309 of 10,711 facts remain on T-REx and 1,561 of 8,915 on CounterFact. Backbone recitation BBR drops by 11.4 and 17.2 percentage points respectively, to 69.7% and 44.3%. The backbone still recites most of these facts with source memory disabled entirely, even though they are the facts their own source supports most clearly. Surviving-memory interference SMI roughly doubles on T-REx and rises by about 60% on CounterFact. Our read on this is that the facts their source supports most clearly are also the facts where surviving memory most often overrides a correct answer, which is why SMI rises as the cohort shrinks.

D Deletion Rules

D.1 Co-LMLM

Let $\hat{q}_f$ denote the query representation captured for fact f under FULL, and let $m_f^\star$ denote the entry it retrieves. We consider three basic deletion rules, each of which includes $m_f^\star$ as an oracle seed:

$$\begin{aligned}\mathcal{R}_\rho^{\mathrm{geo}}(f) &= \{m_f^\star\} \cup \{(k,v) \in \mathcal{M} : \langle \hat{q}_f, k\rangle \geq \rho\},\\ \mathcal{R}^{\mathrm{val}}(f) &= \{m_f^\star\} \cup \{(k,v) \in \mathcal{M} : \mathrm{match}(v, a_f)\},\\ \mathcal{R}^{\mathrm{prov}}(f) &= \{m_f^\star\} \cup \{(k,v) \in \mathcal{M} : \mathrm{src}(v) = \mathrm{src}(m_f^\star)\}.\end{aligned} \tag{6}$$

Here, ρ is a cosine-similarity threshold and $\mathrm{src}(v)$ is the source document of entry v. The geometric rule removes entries close to the query, the value rule removes entries that mention the answer, and the provenance rule removes entries from the same source document as $m_f^\star$.

The <FACT-q> rule. A released question generator receives the answer-mentioning entry selected under FULL; the audit prompt itself is not shown. Each generated question is passed through Co-LMLM's <FACT-q> interface, and entries retrieved above threshold τ are added to the deletion set. This reduces dependence on the particular audit prompt, although initialization still requires an answer-mentioning FULL entry with a locatable answer span.

D.2 NULLs

Let $s_f \in \mathcal{M}$ denote the source associated with fact f. Let $e(\cdot)$ denote the external text encoder, and let $\mathcal{N}_k(f)$ denote the k sources other than s_f with the highest cosine similarity $\langle e(s), e(s_f)\rangle$. We consider three basic deletion rules:

$$\begin{aligned} \mathcal{R}^{\text{prov}}(f) &= \{s_f\}, \\ \mathcal{R}^{\text{geo}}_k(f) &= \{s_f\} \cup \mathcal{N}_k(f), \\ \mathcal{R}^{\text{val}}(f) &= \{s_f\} \cup \{s \in \mathcal{N}_K(f) : \text{match}(\text{text}(s), a_f)\}. \end{aligned} \tag{7}$$

The provenance rule removes only the associated source, the geometric rule adds the k nearest source articles, and the value rule adds, among the $K = 500$ nearest sources, those whose article text mentions the answer. Their intersection, $\mathcal{R}^{\text{geo}}_k \cap \mathcal{R}^{\text{val}}$, adds only the sources among the k nearest that mention the answer.

Routing under DEL-ON. Disabling a deletion set $\mathcal{R}(f)$ zeroes every sink neuron in the union of the masks μ_s, $s \in \mathcal{R}(f)$. NULLs then activates the mask of the next-closest source in its routing list for s_f that is not in $\mathcal{R}(f)$. Neurons shared with a deleted mask stay zeroed, so the routed source cannot reactivate them.

E Co-LMLM-Specific Analyses

This appendix defines the Co-LMLM-specific tests summarized in Section 4.6 and reports their full results. Throughout, $\hat{q}_f$ is the query formed for x_f under FULL, $\mathcal{E}_K(f)$ is the set of K memory entries nearest to $\hat{q}_f$, and match is the answer-matching rule from Section 4.1.

The radius sweep and re-insertion test use the audited cohort $\mathcal{F}$. The answer-density and probe analyses instead use every fact that passes the pre-deletion check, including facts that Co-LMLM does not recite under FULL. These sets contain $8{,}258$ facts on T-REx, $1{,}001$ on PopQA, 633 on Google-RE, $4{,}085$ on CounterFact, and $3{,}754$ on ZsRE. Backbone recitation in Figures 3 and 4 is therefore computed over a larger set and differs slightly from BBR in Table 1; on T-REx, for example, the rates are 25.0% and 25.6%.

E.1 Answer density

We measure how densely an answer appears around a query in the memory store:

$$\text{dens}(f) = \big|\{(k, v) \in \mathcal{E}_K(f) : \text{match}(v, a_f)\}\big|. \tag{8}$$

We use $K = 500$, so the measure is right-censored at 500. Larger values mean that more entries near the query mention the answer.

Figure 3 asks whether facts whose answers appear more often near the query are also more likely to be recited by the backbone. The association is positive on every prompt set, but weaker for Co-LMLM than for the reference models, with Spearman correlations of 0.05–0.27 against 0.21–0.63. The gap remains in the densest quartile: on T-REx, the Co-LMLM backbone recites 43.8% of these facts, compared with 72.0% for SmolLM2-360M and 80.2% for Std-LM-360M. Frequent mention of an answer in the external memory therefore does not imply that the backbone recites it.

E.2 Representation probe

Backbone recitation records whether Co-LMLM generates the answer without retrieval, but not whether information about the answer is absent from its internal representations. We therefore test whether the answer can be decoded from $\hat{q}_f$, taken from the first retrieval event of the FULL trace.

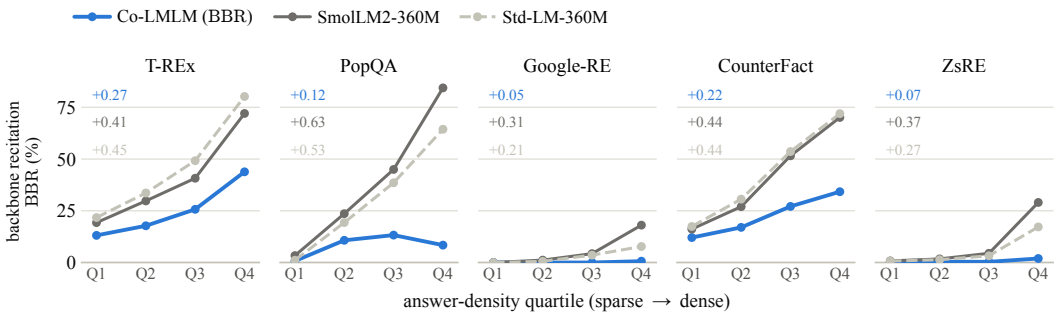


Figure 3: Backbone recitation BBR by answer-density quartile, from Q1 (sparsest) to Q4 (densest), with dataset-specific quartile edges. Density counts answer-mentioning entries among the 500 entries nearest the query. The reference models have no separable memory, so their curves give accuracy on the same facts. Numbers in each panel are Spearman rank correlations between density and recitation.

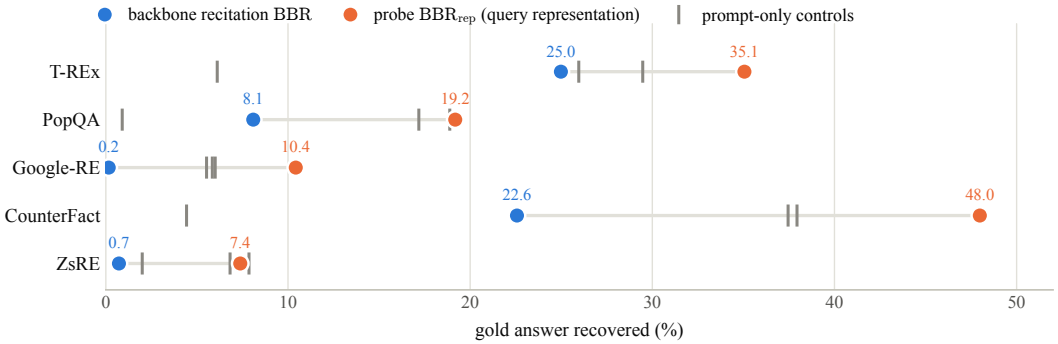


Figure 4: Linear-probe recitation $\mathrm{BBR}_{\mathrm{rep}}$ from Co-LMLM query representations, compared with backbone recitation BBR over all facts that pass the pre-deletion check. The controls use only the prompt or the answer distribution and do not see the query representation.

We fit a linear readout W with ridge regression under five-fold cross-validation. Each held-out representation is scored against a closed candidate set $\mathcal{Y}$ containing the deduplicated gold answers of the probed facts, with candidates represented by hashed character-trigram features $\psi(\cdot)$. We count the probe as correct on f if its top-ranked candidate matches the gold answer:

$$\mathrm{BBR}_{\mathrm{rep}}(f) = \mathbf{1}\left[\mathrm{match}\left(\arg\max_{y\in\mathcal{Y}} \psi(y)^{\top} W \hat{q}_f,\ a_f\right)\right]. \tag{9}$$

To keep the same proposition out of both training and test folds, we group T-REx, Google-RE, CounterFact, and ZsRE by a canonicalized (subject, relation, answer) key. PopQA lacks subject and relation metadata, so it uses fact-disjoint folds.

Probe accuracy may reflect regularities in the prompt or answer distribution rather than information in $\hat{q}_f$. We therefore compare $\mathrm{BBR}_{\mathrm{rep}}$ with three controls that do not see the query representation: hashed character trigrams of the prompt, the mean SmolLM2-360M input-token embedding of the prompt, and a prior that always predicts the most frequent gold answer. We report the margin over the strongest control and the Matthews correlation ϕ between $\mathrm{BBR}_{\mathrm{rep}}(f)$ and $\mathrm{BBR}(f)$.

On T-REx and CounterFact, the probe recovers the gold answer for 35.1% and 48.0% of facts, compared with backbone recitation of 25.0% and 22.6%, and exceeds the strongest control by 5.6 and 10.0 points. On Google-RE, the margin is 4.4 points. The evidence is weak on PopQA, with a margin of 0.3 points, and absent on ZsRE. Per-fact agreement between the probe and the backbone is low on every prompt set ($\phi \leq 0.24$). On several prompt sets, answer information can therefore be decoded from the query representation even when the backbone does not recite the answer.

Table 5: Adversarial re-insertion against the geometric deletion set $\mathcal{R}^{\mathrm{geo}}_{0.85}(f)$, in percent of $\mathcal{F}$. $\mathrm{rec}^{\mathrm{geo}}$ is recitation after deletion and $\mathrm{rec}^{\mathrm{adv}}$ is recitation after inserting one entry at offset ε outside the boundary. Gain^* is the share of facts not recited after deletion that the entry restores, and Reg^* is the share of facts still recited after deletion that are no longer recited after insertion.

			$\varepsilon = 0.01$			$\varepsilon = 0.02$			$\varepsilon = 0.05$		
Dataset	Template	$\mathrm{rec}^{\mathrm{geo}}$	$\mathrm{rec}^{\mathrm{adv}}$	Gain^*	Reg^*	$\mathrm{rec}^{\mathrm{adv}}$	Gain^*	Reg^*	$\mathrm{rec}^{\mathrm{adv}}$	Gain^*	Reg^*
T-REx	verbatim	47.2	66.6	56.2	21.7	66.0	52.6	18.9	65.1	44.8	12.1
	hyphenated		13.7	0.3	71.2	18.3	0.3	61.6	28.3	0.3	40.3
	letter-spaced		13.3	0.0	71.8	17.9	0.0	62.1	28.0	0.0	40.6
	prefix-cue		13.3	0.0	71.8	17.9	0.0	62.1	28.0	0.0	40.6
PopQA	verbatim	62.5	72.4	37.9	6.9	72.4	35.1	5.3	70.8	26.7	2.7
	hyphenated		49.7	1.4	21.4	52.6	1.1	16.6	58.1	0.7	7.6
	letter-spaced		48.8	0.0	22.1	52.0	0.0	16.8	57.8	0.0	7.6
	prefix-cue		48.8	0.0	22.1	52.0	0.0	16.8	57.8	0.0	7.6
Google-RE	verbatim	12.0	82.4	83.0	21.9	80.8	80.9	20.5	77.1	76.1	15.1
	hyphenated		1.0	0.0	91.8	2.0	0.0	83.6	4.3	0.0	64.4
	letter-spaced		1.0	0.0	91.8	2.0	0.0	83.6	4.3	0.0	64.4
	prefix-cue		1.0	0.0	91.8	2.0	0.0	83.6	4.3	0.0	64.4
CounterFact	verbatim	59.0	79.4	68.2	12.8	78.4	64.8	12.2	75.7	55.1	10.0
	hyphenated		17.1	0.0	71.0	21.8	0.0	63.0	32.7	0.0	44.6
	letter-spaced		17.1	0.0	71.0	21.8	0.0	63.0	32.7	0.0	44.6
	prefix-cue		17.1	0.0	71.0	21.8	0.0	63.0	32.7	0.0	44.6
ZsRE	verbatim	41.4	67.6	62.9	25.8	68.3	61.0	21.3	68.9	56.2	13.0
	hyphenated		12.4	3.3	74.8	16.1	3.3	65.8	25.0	2.9	43.9
	letter-spaced		10.3	0.9	76.5	14.1	0.9	67.3	23.2	0.7	44.8
	prefix-cue		9.6	0.0	76.8	13.4	0.0	67.6	22.7	0.0	45.0

E.3 Adversarial re-insertion

A geometric deletion can stop the model from reciting a fact without showing that the fact is robustly removed, since an answer-bearing entry may lie just outside the deletion set and still be retrieved. We test this under a database-administrator threat model, where the adversary knows the deletion radius and can insert a single entry.

For a target fact f, we attack the geometric deletion set $\mathcal{R}^{\mathrm{geo}}_{\rho}(f)$ at $\rho = 0.85$. The adversary inserts an entry (k_a, v_a) whose key lies just outside the boundary:

$$k_a = \gamma \hat{q}_f + \sqrt{1-\gamma^2}\, u, \qquad \gamma = \rho - \varepsilon,$$

where u is a random unit vector orthogonal to $\hat{q}_f$. Then $\langle \hat{q}_f, k_a \rangle = \gamma < \rho$, so the entry survives the deletion while remaining close to the query. The value v_a either states a_f verbatim or uses one of three templates designed to evade match: hyphenated (each word split by a hyphen), letter-spaced (the answer spelled letter by letter), or prefix-cue (the first and last characters of the answer and its length).

Let $\mathrm{rec}^{\mathrm{geo}}_{\rho}(f) = \mathrm{rec}\big(f; \mathcal{M} \setminus \mathcal{R}^{\mathrm{geo}}_{\rho}(f)\big)$ be recitation after deletion, and

$$\mathrm{rec}^{\mathrm{adv}}_{\rho,\varepsilon}(f) = \mathrm{rec}\big(f; \big(\mathcal{M} \setminus \mathcal{R}^{\mathrm{geo}}_{\rho}(f)\big) \cup \{(k_a, v_a)\}\big) \tag{10}$$

recitation after the attack. Since post-attack recitation alone does not separate facts restored by the attack from facts that survived deletion, we measure per-fact flips:

$$\mathrm{Gain}(f) = \big(1 - \mathrm{rec}^{\mathrm{geo}}_{\rho}(f)\big)\, \mathrm{rec}^{\mathrm{adv}}_{\rho,\varepsilon}(f), \qquad \mathrm{Reg}(f) = \mathrm{rec}^{\mathrm{geo}}_{\rho}(f)\, \big(1 - \mathrm{rec}^{\mathrm{adv}}_{\rho,\varepsilon}(f)\big). \tag{11}$$

Both are averaged over $\mathcal{F}$, and we report the conditional forms

$$\mathrm{Gain}^* = \frac{\mathrm{Gain}}{1 - \mathrm{rec}^{\mathrm{geo}}_{\rho}}, \qquad \mathrm{Reg}^* = \frac{\mathrm{Reg}}{\mathrm{rec}^{\mathrm{geo}}_{\rho}},$$

the share of facts not recited after deletion that the attack restores, and the share of facts still recited after deletion that are no longer recited after insertion.

A single entry that states the answer verbatim reverses a large fraction of geometric deletions. At $\varepsilon = 0.01$, Gain^* ranges from 37.9% on PopQA to 83.0% on Google-RE. The three evading templates restore almost nothing, at most 3.3%, but they still win retrieval and cause many facts that were

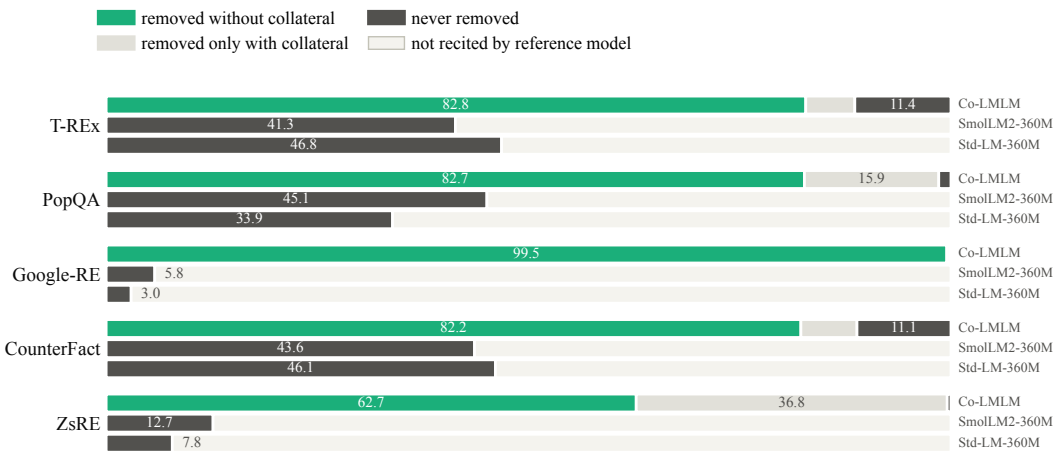


Figure 5: Best outcome for each audited fact across the Co-LMLM geometric radius sweep. "Removed without collateral" ($G(f) = 0$) means that some radius makes the target no longer recited without affecting a measured control fact. "Removed only with collateral" means that every radius that makes the target no longer recited also affects at least one measured control fact. "Never removed" means that the target remains recited at every swept radius. The reference models have no separable memory; their dark bar is the share of the same facts they recite.

Table 6: Radius sweep split by backbone recitation. For facts with $\mathrm{BBR}(f) = 1$ and $\mathrm{BBR}(f) = 0$, the table gives their number and the share that is no longer recited at some swept radius. For $\mathrm{BBR}(f) = 1$, it also gives the share with $G(f) = 0$.

		$\mathrm{BBR}(f) = 1$			$\mathrm{BBR}(f) = 0$	
Dataset	$\lvert\mathcal{F}\rvert$	n	removed at some ρ	$G(f) = 0$	n	removed at some ρ
T-REx	7,407	1,898	55.9	49.5	5,509	99.9
PopQA	761	64	84.4	53.1	697	99.9
Google-RE	608	1	0.0	0.0	607	100.0
CounterFact	3,479	843	54.5	47.2	2,636	99.9
ZsRE	2,260	20	70.0	20.0	2,240	100.0

recited after deletion to no longer be recited, with Reg^* between 21.4% and 91.8%. Their rates are nearly identical across templates, so this disruption comes from winning retrieval rather than from what the entry says. Moving the entry farther from the boundary weakens both effects, since it is retrieved less often: on T-REx, verbatim Gain^* falls from 56.2% to 44.8% between $\varepsilon = 0.01$ and 0.05. The verbatim entry mentions a_f and would be removed by the value rule, but the evading templates are designed to escape it, and the value rule itself raises surviving-memory interference (Table 12).

E.4 Radius sweep

Figure 2 reports target removal E and collateral damage X (5) averaged over $\mathcal{F}$. To summarize the sweep per fact, we use

$$G(f) = \min_{\rho} \left[(1 - E(f, \rho)) + X(f, \rho) \right], \tag{12}$$

where ρ ranges over the six swept radii. $G(f) = 0$ means that some radius makes the target no longer recited without affecting any measured control fact.

On four prompt sets, 82–99% of targets have some radius that makes them no longer recited without measured collateral damage; on ZsRE the corresponding rate is 62.7% (Figure 5). This does not show that the fact has been removed from the backbone, because E only records whether the model recites the fact under DEL-ON. Table 6 makes this distinction explicit. Among facts that the backbone

recites without any memory, some swept radius still suppresses recitation for 55.9% on T-REx, 54.5% on CounterFact, and 84.4% on PopQA. Geometric deletion can therefore appear successful because surviving retrieval suppresses an answer that the backbone still recites.

F NULLs-Specific Analyses

F.1 Source attribution

Ghosal et al. (2026) argue that because source contributions remain disentangled, the effect of an individual source can be measured by comparing NULLs with that source's sink active and inactive. We measure this effect by comparing the mask of the associated source (Section 3.2) with the mask of an unrelated source (placebo source). NULLs instead uses the next-closest source as the inactive condition, but because that source is selected for its similarity to the associated source, it may cover the same fact.

Let s_f denote the associated source of fact f and $\tilde{s}_f$ its placebo source, and let $\mathcal{F}_{\text{map}}$ denote the facts for which s_f is a source of NULLs. For each fact, we compare how likely the model finds the gold answer a_f under the two masks. Let $\ell(f; s)$ measure, as a natural logarithm, how likely the model is to answer a_f when the mask of source s is active. We define

$$\delta(f) = \ell(f; s_f) - \ell(f; \tilde{s}_f)\,, \qquad \text{Acc}_{\text{attr}} = \frac{1}{|\mathcal{F}_{\text{map}}|} \sum_{f \in \mathcal{F}_{\text{map}}} \mathbf{1}[\delta(f) > 0]\,. \tag{13}$$

If the associated source mask carries no information about the fact, δ is symmetric around zero and $\text{Acc}_{\text{attr}} = 1/2$. A low Acc_{attr} does not necessarily contradict NULLs, which is designed so that facts appearing in many sources are held by the shared backbone rather than by any single sink (Ghosal et al., 2026).

The results from this analysis are reported in Figure 6 and Table 7.

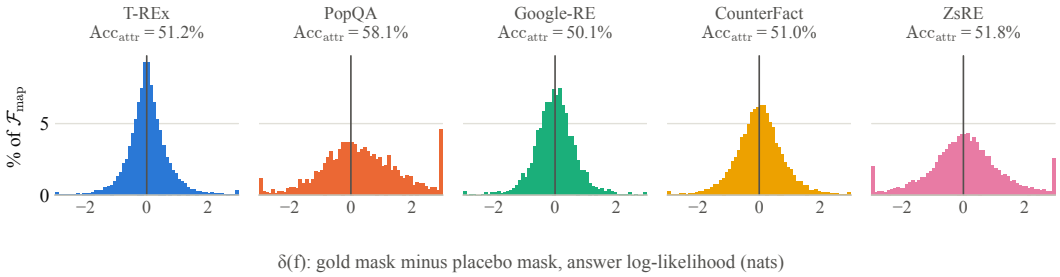


Figure 6: Distribution of $\delta(f)$ over $\mathcal{F}_{\text{map}}$ for each prompt set. Positive values mean the associated source mask makes the answer more likely than the placebo mask, and the vertical line marks $\delta(f) = 0$. Acc_{attr} is the share of facts with $\delta(f) > 0$. Values beyond ± 3 nats are collected in the outer bins.

Table 7: Source attribution over $\mathcal{F}_{\text{map}}$. Acc_{attr} is in percent with Wilson 95% intervals (16), and the median of δ is in nats.

	n	Acc_{attr}	median δ
T-REx	20,938	51.2 [50.5, 51.8]	0.010
PopQA	2,181	58.1 [56.0, 60.1]	0.227
Google-RE	2,480	50.1 [48.1, 52.0]	0.003
CounterFact	17,484	51.0 [50.2, 51.7]	0.016
ZsRE	17,175	51.8 [51.1, 52.6]	0.043

On four of the five prompt sets, the associated source mask makes the answer more likely for only 50–52% of facts, close to the $1/2$ expected if it carried no information. The median gain is at most

0.043 nats, meaning the associated source mask makes the answer at most about 4% more likely than the placebo mask. Only PopQA shows a clearer effect, at 58.1%. Most facts in our prompt sets therefore do not appear to be specific to their source article. This is consistent with Ghosal et al. (2026), who state that information shared across sources is stored in a shared backbone. As a consequence, post-deletion recitation PDR and backbone recitation BBR both depend mainly on what the shared backbone can answer on its own, which explains why they are similar in Tables 1 and 11. Since the pre-deletion check keeps exactly the facts with $\delta(f) > 0$, the facts that pass it are also close to the half of $\mathcal{F}_{\text{map}}$ that falls on the positive side by chance, which Appendix C addresses by tightening the margin.

F.2 Effect of the associated source

Source attribution asks whether the associated source mask makes the answer more likely. We now ask whether it changes the answer the model generates. For each fact, we compare FULL, where only the associated source mask is active, with DEL-OFF, where no source mask is active. We define:

$$\begin{aligned} \text{Gain}_{\text{own}}(f) &= \text{rec}_{\text{FULL}}(f)\,(1 - \text{rec}_{\text{DEL-OFF}}(f))\,, \\ \text{Loss}_{\text{own}}(f) &= (1 - \text{rec}_{\text{FULL}}(f))\,\text{rec}_{\text{DEL-OFF}}(f). \end{aligned} \tag{14}$$

Gain_{own} marks facts the model recites only with the associated source mask, and Loss_{own} marks facts it recites only without it. Reporting both rates, rather than only the difference in accuracy between FULL and DEL-OFF, shows whether the associated source mask makes many answers correct and many answers incorrect, or changes few answers at all.

We compute both rates over all facts that pass the pre-deletion check, since the audited cohort $\mathcal{F}$ excludes every case of Loss_{own}, and compare them with McNemar's exact test (17). Table 8 reports the results.

Table 8: Effect of the associated source over all facts that pass the pre-deletion check, in percent with Wilson 95% intervals (16). Gain_{own} is recited under FULL only and Loss_{own} is recited under DEL-OFF only. p is McNemar's exact test (17) of $\text{Gain}_{\text{own}} = \text{Loss}_{\text{own}}$.

	n	Gain_{own}	Loss_{own}	p
T-REx	10,711	4.7 [4.3, 5.1]	3.4 [3.1, 3.8]	4.8×10^{-6}
PopQA	1,268	0.5 [0.2, 1.0]	2.1 [1.4, 3.0]	5.4×10^{-4}
Google-RE	1,242	1.0 [0.6, 1.8]	0.7 [0.4, 1.4]	0.523
CounterFact	8,915	3.4 [3.1, 3.8]	1.6 [1.3, 1.8]	3.2×10^{-15}
ZsRE	8,901	0.6 [0.4, 0.7]	0.5 [0.4, 0.7]	0.679

Both Gain_{own} and Loss_{own} stay below 5% on every prompt set. This agrees with the source attribution results, where the associated source mask barely changes how likely the answer is.

F.3 Next-closest versus placebo source

After the associated source is deleted, NULLs activates the next-closest source instead (Ghosal et al., 2026). If the model still recites the fact, there are two possible reasons: the next-closest source also holds the fact, or having any source mask active is enough. To tell these apart, we repeat DEL-ON with the next-closest source replaced by the placebo source $\tilde{s}_f$. The placebo is chosen from articles that are unrelated to s_f, and is therefore unlikely to contain the same fact. If the next-closest source holds the fact, the model should recite it less often with the placebo. Let $\text{rec}_{\text{plc}}(f)$ be recitation under this replacement. We compute surviving-memory support SMS and surviving-memory interference SMI as in (3), with rec_{plc} in place of $\text{rec}_{\text{DEL-ON}}$:

$$\text{SMS}_{\text{plc}}(f) = \text{rec}_{\text{plc}}(f)\,(1 - \text{rec}_{\text{DEL-OFF}}(f))\,, \qquad \text{SMI}_{\text{plc}}(f) = (1 - \text{rec}_{\text{plc}}(f))\,\text{rec}_{\text{DEL-OFF}}(f), \tag{15}$$

and compare the two versions of DEL-ON with McNemar's exact test (17). Table 9 reports the results.

Table 9: SMS and SMI over the audited cohort $\mathcal{F}$ with the next-closest source (DEL-ON) or a placebo source unrelated to the associated source active after deletion, in percent with Wilson 95% intervals (16). Both are compared with sinks-zero DEL-OFF. p is McNemar's exact test (17) between the two conditions.

	n		Next-closest	Placebo	p
T-REx	2,669	SMS	8.1 [7.1, 9.1]	7.7 [6.7, 8.8]	0.284
		SMI	8.4 [7.4, 9.5]	8.9 [7.9, 10.1]	
PopQA	22	SMS	13.6 [4.7, 33.3]	0.0 [0.0, 14.9]	0.021
		SMI	13.6 [4.7, 33.3]	36.4 [19.7, 57.0]	
Google-RE	35	SMS	8.6 [3.0, 22.4]	17.1 [8.1, 32.7]	1.000
		SMI	17.1 [8.1, 32.7]	22.9 [12.1, 39.0]	
CounterFact	791	SMS	17.3 [14.8, 20.1]	16.9 [14.5, 19.7]	0.840
		SMI	11.8 [9.7, 14.2]	10.9 [8.9, 13.2]	
ZsRE	79	SMS	22.8 [14.9, 33.2]	19.0 [11.9, 29.0]	0.851
		SMI	20.3 [12.9, 30.4]	19.0 [11.9, 29.0]	

Replacing the next-closest source with the placebo leaves surviving-memory support SMS and surviving-memory interference SMI largely unchanged, and McNemar's test finds no significant difference on any prompt set except PopQA, whose cohort contains only 22 facts. SMS and SMI in NULLs therefore appear to depend little on which source remains active after deletion.

In short, the three analyses are consistent with the design described by Ghosal et al. (2026). The facts in our prompt sets appear to be held mainly by the shared backbone rather than by individual source sinks. For such facts, deleting a source leaves the answer largely unchanged. Whether the model still recites a fact after deletion depends mostly on the shared backbone, not on the deleted or the remaining source.

G Additional Results

Additional audit outcomes. Table 10 gives the full results behind Table 1, including FULL accuracy and confidence intervals. Table 11 reports the same outcomes for PopQA and ZsRE.

Table 10: Full version of Table 1: outcomes under each system's standard deletion rule (quantities defined in Section 3.1), in percent of the audited cohort $\mathcal{F}$, $n = |\mathcal{F}|$. FULL is accuracy over the facts that pass the pre-deletion check (Section 4.1). Reference models have no separable memory. Intervals are Wilson 95% intervals (16). PopQA and ZsRE are in Table 11.

	T-REx	Google-RE	CounterFact
Co-LMLM			
n	7,407	608	3,479
FULL	89.7 [89.0, 90.3]	96.1 [94.2, 97.3]	85.2 [84.0, 86.2]
Post-deletion recitation PDR	8.3 [7.7, 9.0]	0.3 [0.1, 1.2]	5.9 [5.1, 6.7]
Backbone recitation BBR	25.6 [24.6, 26.6]	0.2 [0.0, 0.9]	24.2 [22.8, 25.7]
Surviving-memory support SMS	1.5 [1.2, 1.8]	0.2 [0.0, 0.9]	1.6 [1.2, 2.1]
Surviving-memory interference SMI	18.8 [17.9, 19.7]	0.0 [0.0, 0.6]	19.9 [18.7, 21.3]
Spurious-deletion rate Φ	20.5 [19.5, 21.4]	0.0 [0.0, 0.6]	21.2 [19.8, 22.6]
SmolLM2-360M	41.3 [40.2, 42.5]	5.8 [4.2, 7.9]	43.6 [42.0, 45.3]
Std-LM-360M	46.8 [45.7, 47.9]	3.0 [1.9, 4.6]	46.1 [44.4, 47.7]
NULLs			
n	2,669	35	791
FULL	24.9 [24.1, 25.7]	2.8 [2.0, 3.9]	8.9 [8.3, 9.5]
Post-deletion recitation PDR	80.7 [79.2, 82.2]	54.3 [38.2, 69.5]	67.1 [63.8, 70.3]
Backbone recitation BBR	81.1 [79.5, 82.5]	62.9 [46.3, 76.8]	61.6 [58.1, 64.9]
Surviving-memory support SMS	8.1 [7.1, 9.1]	8.6 [3.0, 22.4]	17.3 [14.8, 20.1]
Surviving-memory interference SMI	8.4 [7.4, 9.5]	17.1 [8.1, 32.7]	11.8 [9.7, 14.2]
Spurious-deletion rate Φ	43.7 [39.5, 48.0]	37.5 [18.5, 61.4]	35.8 [30.2, 41.8]
SmolLM2-1.7B	82.2 [80.7, 83.6]	40.0 [25.6, 56.4]	64.7 [61.3, 68.0]

Table 11: PopQA and ZsRE version of Table 10: outcomes under each system's standard deletion rule (quantities defined in Section 3.1), in percent of the audited cohort $\mathcal{F}$, $n = |\mathcal{F}|$. FULL is accuracy over the facts that pass the pre-deletion check (Section 4.1). Reference models have no separable memory. Intervals are Wilson 95% intervals (16).

	PopQA	ZsRE
Co-LMLM		
n	761	2,260
FULL	76.0 [73.3, 78.6]	60.2 [58.6, 61.8]
Post-deletion recitation PDR	1.2 [0.6, 2.2]	0.8 [0.5, 1.3]
Backbone recitation BBR	8.4 [6.6, 10.6]	0.9 [0.6, 1.4]
Surviving-memory support SMS	0.9 [0.4, 1.9]	0.7 [0.4, 1.1]
Surviving-memory interference SMI	8.1 [6.4, 10.3]	0.7 [0.4, 1.1]
Spurious-deletion rate Φ	8.2 [6.5, 10.4]	0.7 [0.4, 1.2]
SmolLM2-360M	45.1 [41.6, 48.6]	12.7 [11.3, 14.1]
Std-LM-360M	33.9 [30.6, 37.3]	7.8 [6.8, 9.0]
NULLs		
n	22	79
FULL	1.7 [1.1, 2.6]	0.9 [0.7, 1.1]
Post-deletion recitation PDR	72.7 [51.8, 86.8]	40.5 [30.4, 51.5]
Backbone recitation BBR	72.7 [51.8, 86.8]	38.0 [28.1, 49.0]
Surviving-memory support SMS	13.6 [4.7, 33.3]	22.8 [14.9, 33.2]
Surviving-memory interference SMI	13.6 [4.7, 33.3]	20.3 [12.9, 30.4]
Spurious-deletion rate Φ	50.0 [18.8, 81.2]	34.0 [22.2, 48.3]
SmolLM2-1.7B	68.2 [47.3, 83.6]	63.3 [52.3, 73.1]

On PopQA and ZsRE, Co-LMLM again has little recitation supported by surviving memory under the standard deletion rule. On PopQA, surviving-memory interference is much larger than surviving-memory support, so deletion often makes a fact unrecited even though the backbone can still recite it. On ZsRE, both rates are below 1%. For NULLs, post-deletion recitation and backbone recitation remain close on both prompt sets, consistent with many of these facts being recited by the shared backbone rather than by a particular source memory. The NULLs cohorts are small, however, with only 22 audited facts on PopQA and 79 on ZsRE, so these rates have wide confidence intervals and should be interpreted cautiously.

Deletion-rule results. Tables 12 and 13 compare the deletion rules using surviving-memory support and surviving-memory interference. The first measures how often the fact is still recited because supporting memory remains outside the deletion set; the second measures how often the surviving memory suppresses an answer that the backbone can recite.

Table 12: Surviving-memory support SMS and surviving-memory interference SMI (3) by deletion rule, in percent of $\mathcal{F}$, with Wilson 95% intervals (16). Each cell is SMS / SMI. The standard rules for Co-LMLM and NULLs are marked with † and ‡, respectively. PopQA and ZsRE are reported in Table 13.

	T-REx	Google-RE	CounterFact
Co-LMLM			
Oracle seed	36.3 / 6.7	14.1 / 0.0	46.0 / 5.7
$\{m_f^\star\}$	[35.2, 37.4] / [6.2, 7.3]	[11.6, 17.1] / [0.0, 0.6]	[44.4, 47.7] / [4.9, 6.5]
Geometric	30.0 / 8.4	11.8 / 0.0	41.7 / 7.0
$\mathcal{R}^{\mathrm{geo}}_{0.85}$	[28.9, 31.0] / [7.8, 9.1]	[9.5, 14.7] / [0.0, 0.6]	[40.1, 43.4] / [6.2, 7.8]
Provenance	36.2 / 6.7	13.8 / 0.0	46.1 / 5.6
$\mathcal{R}^{\mathrm{prov}}$	[35.1, 37.3] / [6.1, 7.2]	[11.3, 16.8] / [0.0, 0.6]	[44.4, 47.7] / [4.9, 6.4]
<FACT-q>	36.2 / 6.8	14.1 / 0.0	45.9 / 5.7
$\mathcal{R}^{\mathrm{FACT\text{-}q}}$	[35.1, 37.3] / [6.2, 7.4]	[11.6, 17.1] / [0.0, 0.6]	[44.3, 47.6] / [5.0, 6.6]
Value	1.5 / 18.6	0.2 / 0.0	1.5 / 19.9
$\mathcal{R}^{\mathrm{val}}$	[1.2, 1.8] / [17.8, 19.5]	[0.0, 0.9] / [0.0, 0.6]	[1.2, 2.0] / [18.6, 21.2]
Geometric ∪ value†	1.5 / 18.8	0.2 / 0.0	1.6 / 19.9

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Table 12: Table 12 (continued).

	T-REx	Google-RE	CounterFact
$\mathcal{R}^{\mathrm{geo}}_{0.85} \cup \mathcal{R}^{\mathrm{val}}$	[1.2, 1.8] / [17.9, 19.7]	[0.0, 0.9] / [0.0, 0.6]	[1.2, 2.1] / [18.7, 21.3]
Geometric ∪ value ∪ provenance	1.4 / 18.8	0.2 / 0.0	1.6 / 20.0
$\mathcal{R}^{\mathrm{geo}}_{0.85} \cup \mathcal{R}^{\mathrm{val}} \cup \mathcal{R}^{\mathrm{prov}}$	[1.2, 1.7] / [17.9, 19.7]	[0.0, 0.9] / [0.0, 0.6]	[1.2, 2.1] / [18.7, 21.3]
NULLs			
Provenance‡	8.1 / 8.4	8.6 / 17.1	17.3 / 11.8
$\mathcal{R}^{\mathrm{prov}}$	[7.1, 9.1] / [7.4, 9.5]	[3.0, 22.4] / [8.1, 32.7]	[14.8, 20.1] / [9.7, 14.2]
Geometric	8.1 / 8.3	8.6 / 17.1	17.2 / 10.4
$\mathcal{R}^{\mathrm{geo}}_{4}$	[7.2, 9.2] / [7.3, 9.4]	[3.0, 22.4] / [8.1, 32.7]	[14.7, 20.0] / [8.4, 12.7]
Value	7.9 / 4.7	11.4 / 11.4	16.6 / 5.6
$\mathcal{R}^{\mathrm{val}}$	[6.9, 9.0] / [3.9, 5.6]	[4.5, 26.0] / [4.5, 26.0]	[14.1, 19.3] / [4.2, 7.4]
Geometric ∩ value	8.3 / 8.3	11.4 / 14.3	16.9 / 10.0
$\mathcal{R}^{\mathrm{geo}}_{4} \cap \mathcal{R}^{\mathrm{val}}$	[7.3, 9.4] / [7.3, 9.4]	[4.5, 26.0] / [6.3, 29.4]	[14.5, 19.7] / [8.1, 12.3]

Table 13: Surviving-memory support SMS and surviving-memory interference SMI by deletion rule on PopQA and ZsRE, in percent of $\mathcal{F}$, with Wilson 95% intervals (16). Each cell is SMS / SMI. Notation as in Table 12.

	PopQA	ZsRE
Co-LMLM		
Oracle seed	67.3 / 1.2	45.0 / 0.2
$\{m^\star_f\}$	[63.9, 70.5] / [0.6, 2.2]	[42.9, 47.0] / [0.1, 0.5]
Geometric	56.2 / 2.1	40.9 / 0.4
$\mathcal{R}^{\mathrm{geo}}_{0.85}$	[52.7, 59.7] / [1.3, 3.4]	[38.9, 42.9] / [0.2, 0.8]
Provenance	67.4 / 1.2	44.5 / 0.2
$\mathcal{R}^{\mathrm{prov}}$	[64.0, 70.6] / [0.6, 2.2]	[42.4, 46.5] / [0.1, 0.5]
<FACT-q>	65.0 / 1.6	44.7 / 0.2
$\mathcal{R}^{\texttt{FACT-q}}$	[61.6, 68.4] / [0.9, 2.7]	[42.7, 46.8] / [0.1, 0.5]
Value	1.3 / 8.1	0.8 / 0.7
$\mathcal{R}^{\mathrm{val}}$	[0.7, 2.4] / [6.4, 10.3]	[0.5, 1.2] / [0.4, 1.1]
Geometric ∪ value†	0.9 / 8.1	0.7 / 0.7
$\mathcal{R}^{\mathrm{geo}}_{0.85} \cup \mathcal{R}^{\mathrm{val}}$	[0.4, 1.9] / [6.4, 10.3]	[0.4, 1.1] / [0.4, 1.1]
Geometric ∪ value ∪ provenance	0.9 / 8.1	0.7 / 0.7
$\mathcal{R}^{\mathrm{geo}}_{0.85} \cup \mathcal{R}^{\mathrm{val}} \cup \mathcal{R}^{\mathrm{prov}}$	[0.4, 1.9] / [6.4, 10.3]	[0.4, 1.1] / [0.4, 1.1]
NULLs		
Provenance‡	13.6 / 13.6	22.8 / 20.3
$\mathcal{R}^{\mathrm{prov}}$	[4.7, 33.3] / [4.7, 33.3]	[14.9, 33.2] / [12.9, 30.4]
Geometric	13.6 / 13.6	22.8 / 17.7
$\mathcal{R}^{\mathrm{geo}}_{4}$	[4.7, 33.3] / [4.7, 33.3]	[14.9, 33.2] / [10.9, 27.6]
Value	9.1 / 13.6	12.7 / 19.0
$\mathcal{R}^{\mathrm{val}}$	[2.5, 27.8] / [4.7, 33.3]	[7.0, 21.8] / [11.9, 29.0]
Geometric ∩ value	13.6 / 18.2	22.8 / 17.7
$\mathcal{R}^{\mathrm{geo}}_{4} \cap \mathcal{R}^{\mathrm{val}}$	[4.7, 33.3] / [7.3, 38.5]	[14.9, 33.2] / [10.9, 27.6]

For Co-LMLM, PopQA and ZsRE follow the same pattern as the main prompt sets. Rules that do not use the answer text leave substantial surviving-memory support. The value rule, which explicitly uses the answer text to identify entries, and the unions that contain it reduce surviving-memory support to around 1% or below. On PopQA, this reduction is accompanied by more surviving-memory interference; on ZsRE, both rates remain below 1%.

For NULLs, changing the deletion rule has a smaller and less consistent effect. The value rule lowers surviving-memory support on both prompt sets, but combining it with the geometric rule does not consistently reduce it further. These estimates are particularly uncertain on PopQA and ZsRE because the audited cohorts are small.

Additional deletion breadth results. Figure 7 shows the remaining deletion breadth for PopQA and ZsRE. The other prompt sets are shown in Figure 2.

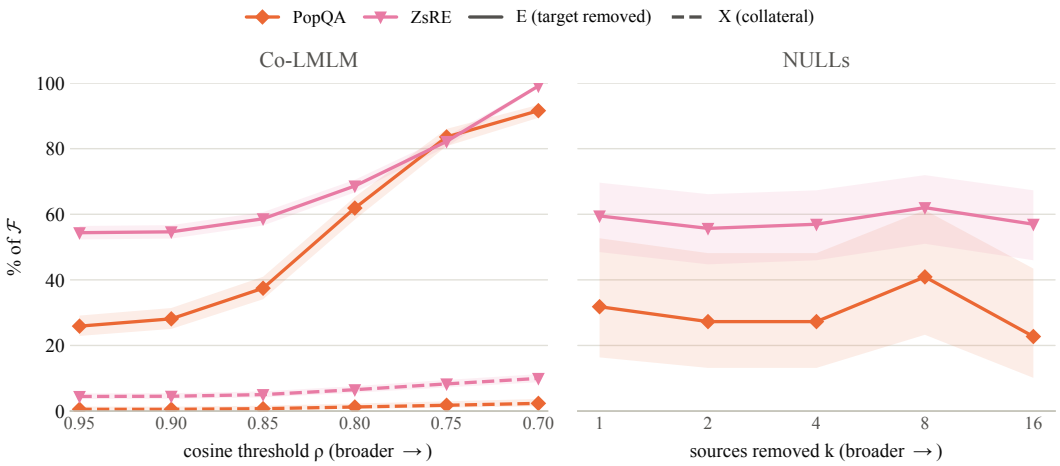


Figure 7: Effect of increasing deletion breadth on PopQA and ZsRE. Quantities and intervals are defined as in Figure 2. For NULLs, both prompt sets had too few control facts for X to be plotted.

For Co-LMLM, broader geometric deletion steadily increases target removal on both PopQA and ZsRE. Measured collateral damage remains much smaller, although it increases at the broadest settings. For NULLs, increasing the number of removed sources does not produce a consistent improvement: target removal fluctuates on PopQA and remains roughly stable on ZsRE, with wide confidence intervals. We do not report collateral damage for these NULLs prompt sets because too few control facts were available. Broader source-level deletion therefore does not reliably remove more targets on either prompt set.

H Audit Configuration

Table 14: Audit configuration. Ranges span the five datasets.

Group	Setting	Value
Shared	Datasets	T-REx, PopQA, Google-RE, CounterFact, ZsRE
	Prompts per dataset	full released sets, 2,766–27,630
	Decoding / max new tokens	greedy / 12
	Global seed	0
Co-LMLM		
Models & index	Audited checkpoint	`lil-lab/CoLMLM-360M-FW`
	Reference models	`HuggingFaceTB/SmolLM2-360M`; `lil-lab/CoLMLM-Standard-LM-Baseline-360M-FW` (scored on the same audited facts; the three conditions coincide, so PDR = BBR and SMS = SMI = 0)
	Memory store	`lil-lab/co-lmlm-360m-fw-fineweb-wiki-index`, ∼2.2B entries (1.07 TB)
	Retrieval threshold τ	0.7
	Retrieval top-k	1
Deletion rule	Standard rule	geometric ∪ value

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Table 14: Audit configuration (continued).

Group	Setting	Value
	Geometric radius ρ	0.85
	Max deletion-set size	10,000
	Deletion rules	oracle seed; geometric; value; provenance; hybrid (geometric ∪ value ∪ provenance); <FACT-q>
<FACT-q>	Question generator	lil-lab/CoLMLM-Question-Generator (LoRA) over Qwen/Qwen2.5-1.5B-Instruct; audit prompts never shown to the generator
	Sampling	1 greedy + 10 sampled (temperature 0.8, top-p 0.95); 5 deduplicated questions kept
	Query embedding	Co-LMLM <FACT-q> head
	Retrieval threshold	0.7
DEL-OFF	Default	null-retrieval
	Sensitivity	forbid-token
Answer density	Envelope / cap	top-500 by cosine to $\hat{q}_f$; right-censored at cap
	Binning	dataset-specific quartiles
Radius sweep	Radii ρ	$\{0.95, 0.90, 0.85, 0.80, 0.75, 0.70\}$
	Neighbor ball (cosine)	0.5
	Neighbor cap / min count	20 / 5
Representation probe	Estimator	ridge regression (ranking)
	Ridge λ	1.0
	Folds	5, proposition-disjoint (fact-disjoint on PopQA)
	Answer features	hashed char n-grams, dim 2,048
	Candidate answers	284–1,960
	Prompt-only controls	prompt char-trigrams (dim 2,048); mean prompt input-token embeddings (SmolLM2-360M input table); gold-answer prior
Adversarial re-insertion	Attacked deletion set	geometric, $\rho = 0.85$
	Offsets ε	$\{0.01, 0.02, 0.05\}$
	Templates	verbatim, hyphenated, letter-spaced, prefix-cue
	Inserted entries	1
NULLs		
Models	Audited checkpoint	gauravrghosal/NULLS-Wikipedia-Full (1B, 6,407,814 sources)
	Reference model	HuggingFaceTB/SmolLM2-1.7B
	Backbone / sink pool neurons	500 / 7,692 per layer (released configuration)
	Sink activation probability	0.013 (∼100 active sinks per source)
	Context length	1,024
Cohort	Source mapping	subject article title, exact or MediaWiki-normalized match
	Verification gate	answer log-likelihood, associated-source mask vs. placebo mask; margin 0
	Placebo source	deterministic; cosine to associated source at most 0.5
	Facts after gate	1,242–10,711
DEL-ON routing	Activated source	next-closest surviving source (from a top-5 list)

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Table 14: Audit configuration (continued).

Group	Setting	Value
	Routing encoder	`sentence-transformers/all-MiniLM-L6-v2` over article titles
	Excluded neurons	union of the masks in the deletion set
Deletion rule	Standard rule	provenance (the associated source)
	Geometric-rule encoder	`sentence-transformers/all-mpnet-base-v2`
	Deletion rules	provenance; geometric ($k = 4$); value (envelope $K = 500$); geometric $\cap$ value ($k = 4$)
	Breadth sweep k	$\{1, 2, 4, 8, 16\}$
DEL-OFF	Default	sinks-zero (backbone only)
	Sensitivity	placebo-sink

I Statistical Uncertainty

For a rate $\hat{p}$ estimated on n facts, we use the Wilson score interval (Wilson, 1927)

$$\frac{1}{1+z^2/n}\left[\hat{p}+\frac{z^2}{2n}\ \pm\ z\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}+\frac{z^2}{4n^2}}\right], \qquad z=z_{1-\alpha/2}=1.96. \tag{16}$$

The interval contains the rates that are consistent with the observed count at the 95% level, and its width decreases with n. It remains inside $[0, 1]$ and does not collapse to zero width near the boundaries.

Paired comparisons on shared facts use McNemar tests (McNemar, 1947). Writing b and c for the discordant counts, the two-sided exact p-value is

$$p=\min\left(1,\ 2\sum_{k=0}^{\min(b,c)}\binom{b+c}{k}2^{-(b+c)}\right). \tag{17}$$

Here, b is the number of facts correct under the first condition only and c the number correct under the second condition only. A small p indicates that the two conditions differ on more facts than would be expected by chance, and facts on which they agree do not enter the test.